Exam

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1 Problem A1

What is the cardinality of the set of all real numbers from $(0, 1)$, whose decimal representation does not contain odd digits?

(a) This set has cardinality of $\mathbb{N}$ (that is $\aleph_0$).

(b) This set has cardinality of $\mathbb{R}$ (that is continuum $\mathfrak{c}$).

(c) This set is finite.

(d) This set has cardinality that is in between $\aleph_0$ and $\mathfrak{c}$.

2 Problem A2

Find $\lim_{n\to\infty} \frac{2^{n+1}}{3^n - 2^n}$.

(a) $\frac{2}{3}$ (b) 1 (c) 0 (d) $+\infty$

3 Problem A3

Find $\gcd(2^{100}, 20^{10})$.

(a) 2^{10} (b) 2^{20} (c) 2^{40} (d) 20^{10}

4 Problem A4

Find the sum of the roots of $z^2 - z + 2i = 0$.

(a) 1 (b) $\sqrt{5}$ (c) 0 (d) $1 + 2i$

5 Problem A5

What is the set $M = \bigcup_{\alpha\in[0,1]} A_\alpha$, if $A_\alpha = \{x : \alpha^2 \leqslant x \leqslant \alpha\}$?

(a) $M = \{0, 1\}$ (b) $M = [0, 1]$ (c) $M = (0, 1)$ (d) $M = [\frac{1}{4}, \frac{1}{2}]$

6 Problem A6

What will the following code output?

```
a = [2]
for i in 1:8
```

```
    if mod(2^i, 9) == 1
        push!(a, i)
    end
end
println(sum(a))
```

(a) 6 (b) 8 (c) 10 (d) this code will return an error.

7 Problem A7

The set A has 2 elements, the set B has 4 elements. How many surjective functions $f : P(A) \times B \to P(A) \times B$ are there?

(a) 6! (b) 8! (c) 12! (d) 16!

8 Problem A8

Choose the statement to which the statement $(P \implies Q) \wedge (P \implies R)$ is equivalent

(a) $(P \wedge Q) \implies R$ (b) $P \implies (Q \vee R)$ (c) $(P \vee Q) \implies R$ (d) $P \implies (Q \wedge R)$

9 Problem A9

Find the coefficient of x^{20} in the expression of $(x^2 - \frac{1}{x})^{100}$.

(a) $\binom{100}{20}$ (b) $\binom{100}{40}$ (c) $-\binom{100}{60}$ (d) $\binom{100}{80}$

10 Problem A10

Consider the set

$$S = \left\{ \frac{m}{n} : m, n \in \mathbb{N},\ m^2 < 2n^2 \right\}.$$

Find $\sup S$ in $\mathbb{R}$ (with the usual order).

(a) $\sqrt{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{2}$ (d) It doesn't exist.

11 Problem B1

Propositions P, Q, and R have the same truth value. Prove that the proposition $(P \vee \neg Q) \wedge (Q \vee \neg R) \wedge (R \vee \neg P)$ is always true.

12 Problem B2

Let R be a relation on $\mathbb{R}$ such that xRy if and only if $x - y \in \mathbb{Q}$. Prove that R is an equivalence relation.

13 Problem B3

Find the number of natural numbers not exceeding 200 that are not divisible by 5 or by 11.

14 Problem B4

The following code aims to find all elements in the array `a_array` that are greater than 13 and are not present in array `b_array`. Fill in the missing bits. (The syntax for and in Julia is &&, for or is ||, and for not is !).

```
a_array = [2, 71, 14, 11, 11, 2, 12, 14, 120]
b_array = [14, 2, 121, 11, 2025]
result_array = []
for a in a_array
    if ...
        push!(....................., ........................)
    end
end
```

15 Problem B5

Depict on a complex plane the set $\{z \in \mathbb{C} : |iz - 1| = 1\}$.

16 Problem B6

Provide an example of strictly decreasing divergent sequence. (No explanations required).

17 Problem B7

For sets A and B prove that $A \setminus B = A \setminus (A \cap B)$.

18 Problem B8

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as $f(x) = \sqrt{|x|}$. Find $f(A)$ if $A = (-\infty, -1) \cup (0, 1)$.

19 Problem B9

Prove that two hyperplanes in $\mathbb{R}^4$ given by

$$2x_1 - 2x_2 + x_3 + 3x_4 = 5 \quad \text{and} \quad x_1 - 3x_2 + x_3 - 3x_4 = 5$$

are orthogonal.

20 Problem B10

Integers a, b are such that $3 \nmid ab$. Prove that $a^2 + b^2$ cannot be a perfect square.

21 Problem C1

Let $A = \{1, 2, \ldots, n + 1\}$. How many strictly increasing sequences of length n can be constructed from the elements of A?

22 Problem C2

Prove the inequality for all nonnegative $x_1, \ldots, x_n$

$$\prod_{k=1}^{n} (1 + x_k) \geqslant 1 + \sum_{k=1}^{n} x_k.$$

23 Problem C3

Find the limit $\lim_{n\to\infty}(\sqrt[4]{n^4 - n^2} - n)$.

24 Problem C4

There are 5 points on the plane with integer coordinates. Prove that the midpoint of one of the segments joining these points also has integer coordinates.

25 Problem C5

Let $f(x) = x^2 + px + q$. Prove that at least one of the numbers $|f(0)|, |f(1)|, |f(-1)|$ is greater or equal $\frac{1}{2}$.

26 Problem C6

Prove that if two sets A and B satisfy $|A| = |B|$, then $|P(A)| = |P(B)|$.

27 Problem C7

Find a polynomial with real coefficients among whose roots are 2 and $1 - i$.

28 Problem C8

How many strings of six lowercase letters from the English alphabet contain the letters a and b? (The formula suffices, no need to calculate the concrete number).